

Translating motivational interviewing for the HPV vaccine into a computable ontology model for automated AI conversational interaction

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ABSTRACT

Human papillomavirus (HPV) vaccinations are lower than expected. To protect the onset of head and neck cancers, innovative strategies to improve the rates are needed. Artificial intelligence may offer some solutions, specifically conversational agents to perform counseling methods. We present our efforts in developing a dialogue model for automating motivational interviewing (MI) to encourage HPV vaccination. We developed a formalized dialogue model for MI using an existing ontology-based framework to manifest a computable representation using OWL2. New utterance classifications were identified along with the ontology that encodes the dialogue model. Our work is available on GitHub under the GPL v.3. We discuss how an ontology-based model of MI can help standardize/formalize MI counseling for HPV vaccine uptake. Our future steps will involve assessing MI fidelity of the ontology model, operationalization, and testing the dialogue model in a simulation with live participants.

*Both authors (NM, MA) contributed equally to this research.

CCS CONCEPTS

• **Information systems** → **Web Ontology Language (OWL); Semantic web description languages**; • **Computing methodologies** → **Ontology engineering; Discourse, dialogue and pragmatics; Planning and scheduling; Intelligent agents**; • **Applied computing** → **Consumer health; Health informatics**.

KEYWORDS

conversational agents, chat bots, human papillomavirus, ontology, dialogue systems, cancer, oral health, patient-provider communication

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1 INTRODUCTION

Human papillomavirus (HPV) is a family of viruses that cause genital warts and several types of cancer, including oropharyngeal, cervical, anal, penile, vulvar, and vaginal. Most initial infections with HPV are asymptomatic and self-limiting, and often go undetected. While the infection is usually cleared by the immune system within two years, HPV can remain dormant in the body for years

[12]. According to the Centers for Disease Control and Prevention, HPV is the most common sexually transmitted infection, and it is estimated that almost every sexually active person will contract HPV at some point in their lifetime if they are not vaccinated [9].

In the United States, the Gardasil 9 vaccine series is used to prevent infection with nine high-risk strains of HPV [9]. This vaccine has been shown to be effectively mitigating head and neck cancers [41], as well as prevent other HPV-related infection and diseases [11]. Additionally, the vaccine is effective for males and females, and studies have demonstrated a protective herd effect when vaccination coverage is over 50% of the population [11, 13].

Despite being safe and effective [8, 39], the HPV vaccine is underutilized in the United States. The Healthy People 2030 goal is to have 80% vaccination among adolescents ages 13 – 15 years old [42]. In 2021, only 58.5% of this population received the vaccine [42]. Commonly reported barriers to adolescents receiving the HPV vaccine include parents' need for more information, parental concerns regarding impact on sexual behavior, a low perceived risk of HPV infection, inadequate access to preventive care, and cost [17]. Interventions for this group primarily need to address parental concerns. Additionally, for those who did not receive the HPV vaccine as an adolescent, CDC guidelines recommend vaccination up to 26 years old. Therefore, addressing barriers for young adults who are unvaccinated is just as important as addressing barriers for parents.

For patients of all ages, including young adults, patient-provider communication has an established, sizeable impact on HPV vaccine uptake. Provider recommendation alone can increase initiation of HPV vaccination from 24% up to 60%, with similar effectiveness across all ages and genders [30]. Provider recommendation also increases follow-through and completion of the HPV vaccine series for these groups [30]. In addition, provider discussion of HPV vaccination – a more in-depth conversation than recommendation alone, shows similar increases in HPV vaccine initiation, follow-through and completion, especially among adolescent and young adult populations [30]. Therefore, we cannot overlook the importance of patient-provider communication in increasing HPV vaccination of young adults. One communication framework that has been utilized in eliciting and maintaining behavior change is motivational interviewing.

Relevance to Oral Health and Dentistry. HPV vaccination is a relevant topic for dental clinicians, as it impacts the risk of a patient developing head and neck cancer. Head and neck cancers encompass a heterogeneous set of malignancies that can arise in the oral cavity, pharynx, larynx, sinuses, and salivary glands. In particular, oropharyngeal cancers are more likely to be due to HPV infection than any other causative agents, unlike oral cavity cancers which are frequently associated with tobacco and alcohol use [10]. An important role of the dental practitioner is to identify oral and oropharyngeal cancers in their patients. Both the American Dental Association and American Academy of Pediatric Dentistry have policy statements supporting the use of HPV vaccination to prevent HPV-related oropharyngeal cancers [2, 3]. Therefore, dental practitioners should emphasize the importance of primary prevention using the HPV vaccine. Dental professionals are in a unique position to educate patients about HPV and its link to oral cancers.

By discussing the HPV vaccine, they can advocate for primary prevention, potentially reducing the prevalence of HPV-associated oral cancers in the future.

1.1 Motivational Interviewing

Motivational interviewing (MI) is a construct developed by Miller and Rollnick [38] to effect behavior change, and over time it has been used for a wide variety of populations and health conditions, supported by a growing body of evidence in the literature. At its core, motivational interviewing relies on an open discussion between practitioner and patient, during which it is the patient who evaluates their goals and their actions. Through the course of the discussion, reveals a cognitive dissonance that their actions do not match their goals, and therefore, their actions must change. The practitioner does not impose a goal on the patient, but instead, guides the patient on a journey to understand their health behaviors and cognitive dissonance to motivate the patient to reach their own expressed goal.

Motivational interviewing has shown promising results in addressing vaccine hesitancy, including HPV vaccination [21, 23, 34, 40, 43]. While the majority of such interventions have been focused on parents, there is the potential for similar efficacy when targeting a young adult population. Additionally, young adult populations have shown success with health behavior interventions that utilize social media and other online platforms to communicate and disseminate information [32]. With the development of artificial intelligence chatbots (i.e. conversational agents), we see the possibility to merge motivational interviewing – a complex, conversational framework – and a technology platform that can alleviate the burden and the time commitment for carrying out such conversations for dentists and physicians.

1.2 Ontologies and Artificial Intelligence

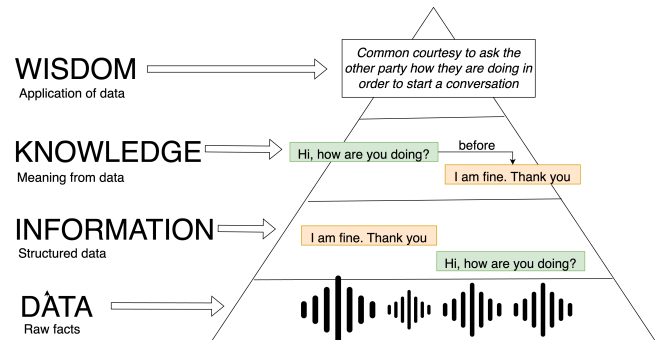


Figure 1: The data, information, knowledge, and wisdom pyramid using speech data as an example.

Ontologies, in the context of computer science, are machine-readable structures that model domain-specific knowledge. Although they have been utilized in various domains for knowledge representation, their significance was notably amplified within the Semantic Web field. They aim to interlink heterogeneous data and information using a structured labeled directed graph network of

vocabularies. Ontologies use an encoded syntax for formal description logic, specifically the Web Ontology Language (OWL2)[16] that is an offshoot of the Resource Description Framework (RDF) [31].

This machine readable syntax have robust semantic encoding to enable machines to preform reasoning and inference, and enables complex sophisticated tasks that go beyond the functions of standard knowledge graphs lacking upper-level semantic definitions [29]. Lastly, the semantics also permits the artifact to comprehensively describe domain knowledge for machines to understand, which gives the machine contextual understanding of the domain.

Researchers describe ontologies as a tool that addresses the knowledge layer [15] of the DIKW (data, information, knowledge, and wisdom) pyramid – a theoretical information theory model that is sometimes attributed to intelligent systems [36]. The DIKW pyramid (Figure 1) describes how data transforms to wisdom. Essentially knowledge enables machines to have awareness and context (i.e., meaning) from data and information. Theoretically, having contextual understanding machines can preform tasks intelligently. In particular with ontologies, these artifacts derive meaning and context with the help of the semantics annotating data and information [29].

Previously, we described the use of ontologies to model the dialogue interaction of machine and humans based off utterance data collected from Wizard of Oz trials (faux conversational agent experiment with live participants)[5, 6]. We theorized that in order for machines/agents to preform dialogue interaction intelligently, it should be aware of the context and meaning of utterance dialogue exchange. This would differ from using statistical or reactive approaches, where the machine do not need to know context.

1.3 Research Objective

This paper focuses on creating a formalized ontology on a structured counseling method - *motivational interviewing*, an evidence-based approach in helping patients self realize their health behaviors and improve their health outcomes. Our focus is on improving HPV vaccine uptake, an area of concern in oral health due to the virus causing head and neck cancers in adulthood. *We propose that an ontological approach can model and describe the method of motivational interviewing for HPV vaccine uptake.* The outcome of this work will provide an ontology that will enable conversational agents to preform some variant of motivational interviewing counseling for HPV vaccine for young adults (18-25; with no children) who missed the opportunity to have the vaccine in their teenage years and encountered in the dental clinic. Beyond the oral health impact, this work could be generalizable and reusable for other machine-based MI approach.

2 METHOD

We created a visualization of our dialogue model (See Section 2.1) as our blueprint for another product we developed - a translational ontology model for motivational interviewing dialogue interaction (See Section 2.2). The work involved planning and segmenting the dialogue into phases (speech acts) to further organize and structure the logical flow of the dialogue model. Our model incorporates

features of MI and other theoretical approaches like Transtheoretical Model of Health Behavior Change (TTM) [33] and Health Belief Model of Behavior Change (HBM) [19]. Details of the theory, methodology, and outputs are described below.

2.1 Design of the Dialogue Model for Motivational Interviewing Toward HPV Vaccine Uptake

2.1.1 Motivational Interviewing Framework. Motivational interviewing (MI) is defined by Miller and Rollnick [24] as “a collaborative, goal-oriented style of communication with particular attention to the language of change. It is designed to strengthen personal motivation and commitment to a specific goal by eliciting and exploring the person’s own reason for change within an atmosphere of acceptance and compassion.” [24] Over time, MI has been used for a wide variety of populations and health conditions and is supported by a growing body of evidence in the literature [38]. MI relies on a flexible and open discussion between the MI practitioner and patient utilizing core counseling skills and four key process: engaging, focusing, evoking and planning using a person-centered approach [24]. During MI, the patient is guided to explore discrepancies between their goals, values and current actions which should build motivation to change [24]. While the majority of MI interventions have been focused on parents, there is the potential for similar efficacy when targeting a young adult population. Additionally, young adult populations have shown success with health behavior interventions that utilize social media and other online platforms to communicate and disseminate information [32]. With the development of artificial intelligence chatbots (i.e. conversational agents) we see the possibility to merge motivational interviewing – a complex, conversational framework – and a technology platform that can alleviate the burden and time commitment of carrying out such conversations for physicians.

A trusted method to elicit this change talk is the OARS approach [35]: *Open ended questions, Affirmations, Reflective listening, and Summarizing*. Unlike the previously outlined acronyms, these statements are meant to be used by the operator. Within the script, the interview phase is governed by an OARS framework. To engage the user in the interview, an open-ended question is asked. The purpose of an open-ended question is to allow the user to drive the conversation and provide the information that is most personally relevant. For example, the first question is, “How do you feel about receiving the HPV vaccine?” Once the user provides a response, an affirmation statement is returned. Continuing our example, the user may say something along the lines of, “I’m nervous to get the vaccine” to which an affirmation statement would be, “It sounds like you have some concerns about the vaccine”. The purpose of using an affirmation statement is to build rapport and express empathy to the user. This will encourage the user to continue speaking, while knowing that the provider agent cares what the user has to say. An additional benefit for the model, is that it provides an opportunity for the user to correct the operator. For example, “No, I don’t really have concerns about the vaccine, just being in a doctor’s office makes me nervous.” This provides additional information to the operator, and will improve response selection as the conversation progresses. After the affirmation statement, the user continues

the discussion, similar to the example given, and then the model provides a listening statement. This statement should reflect back what the user has said. The purpose of the listening statement is to show the user you are listening and understanding, and to once again provide an opportunity for correction. The user will then continue the discussion, after which the model provides a summary statement. These summarizing statements allow the user to track the conversation's progress, and help to transition into the next question or phase of discussion. The majority of the model's interview phase is comprised of this OARS approach (See Appendix Figure 4 for a sample).

Other strategies in MI include asking permission, normalizing, and asking people to evaluate the beneficial and harmful aspects of their behavior [35]. These techniques appear throughout the script (Appendix Figure 4), with dialogue options often beginning with, "If it's alright with you" to request user permission. Additionally, interview questions within the script ask the user to describe both their concerns about HPV vaccination and the benefits of HPV vaccination. Scales can also be used as tools in motivational interviewing through the *importance*, *readiness*, and *confidence* scales [35]. These allow a patient to put into numbers in how important they feel the change is, how ready they feel to change, and how confident they feel in making the change. All these tools can be used by the interviewer to elicit change talk and to reveal cognitive dissonance. Within this script, the *readiness* scale (Appendix Figure 4) is utilized to elicit change talk from the user, and to evaluate where the user may fall on the Transtheoretical Model of behavior change (TTM) stages of change.

2.1.2 Development of Dialogue Model for MI. Using draw.io software, a flowchart script was created to model an artificial intelligence (AI) conversational agent conversation flow utilizing a motivational interviewing framework. The initial framework was developed from the TTM, guiding the conversation from pre-contemplation, to contemplation, to preparation and action stages. Within these stages of change, the content of discussion drew upon the Health Belief Model of behavior change (HBM) to emphasize discussion topics that may best influence behavior change. With this framework and content rooted in public health theory, documented motivational interviewing techniques were applied in the dialogue itself. A breakdown of these theories and their applications within the script is outlined in Appendix Figure 5.

After a brief introduction to the purpose of the conversational agent and engagement of the user, the model can be divided into two main phases: the *interview phase* and the *inform phase*. At the beginning of the *interview phase*, the user is likely in the pre-contemplation or contemplation stages of TTM. This phase utilizes motivational interviewing techniques to prompt discussion and listen to the user, and identifying their feelings, concerns, desires, and readiness regarding HPV vaccination. Possible user responses in this section correspond to aspects of the HBM to best identify personalized targets for behavior change. For example, a user whose responses enter a discussion about whether the vaccine is appropriate for their demographic, has engaged primarily in the construct of perceived susceptibility from the HBM. Later on during the *inform phase* of the model, the provider agent can select information targeting perceived susceptibility.

After the *interview phase*, the user should ideally be in the contemplation or preparation stages of TTM. The purpose of the *inform phase* is to deliver targeted information to the user based on their expressed concerns regarding HPV vaccination. This requires the provider agent to recall the users responses from the *interview phase* in order to select the appropriate information dialogue flow. Concerns posed in the *interview phase* must be addressed in the *inform phase*. The user has the option to express additional concerns or request additional information. The script will continue to offer information only regarding topics the user has expressed an interest in. By the end of the *inform phase*, the user is ideally in the action phase of TTM. A copy of our full documentation of model is available at our GitHub [25] under the GNU General Public License v3 (GPLv3).

2.2 Generating an Ontology-based Representation of our Dialogue Model

2.2.1 PHIDO. The Patient Health Information Dialogue Ontology (PHIDO) is an application ontology that was developed to model the dialogue interaction from a series of Wizard of Oz experiments with potential users, who are decision makers for the HPV vaccine (e.g., parents or young adults) [6]. The ontology represents basic utterances, Searle's speech acts (defined as purposeful utterances [37]), and utterance types that discern the machine (*System Utterance*) and the human participant user (*Participant Utterance*).

The ontology abstracts the utterances as an instance data that forms a network structure, indicative of how dialogue is represented [1, 22]. PHIDO was demonstrated in an actionable artifact for a Java-based dialogue engine that leverages the information contained in the ontology on how to manage and traverse the dialogue interaction structure [5]. The dialogue engine uses some ontology-based machine inferencing to help discern types of utterances for the machine to act upon. For the purpose of the study, we repurposed the ontology as a foundational framework to model and build our motivational interviewing dialogue model for HPV vaccine uptake.

2.2.2 Encoding and Management Process. With each of the utterance described in our model, we produced a spreadsheet that provided an identifier for each possible utterances and their classification. The classification of the utterance was based on the utterance types from the PHIDO. Some of the utterances types were unique to motivational interviewing such as *Reflective Listening*, *Summarizing*, etc (See Appendix Table 1). We created those utterance types and also assigned appropriate standard PHIDO utterances classes - *Participant Utterance* and *System Utterance* (utterance types that clearly delineated speech spoken by the machine or the human participant user). Also, for the new utterance types, we assigned Searle's speech acts. Lastly for each of the utterance types we also noted the connected utterance in the dialogue model. Essentially an utterance that follows a particular utterance(s).

We also developed Java software code using OWL API [18] that imports our spreadsheet data and adds it to the PHIDO ontology. Each of the utterances were treated as instance level data with a connected utterance instance using the *precedes* object property. For example (see Figure 2), where an utterance example is "chained" using the *precedes* object property to another utterance. Also in the

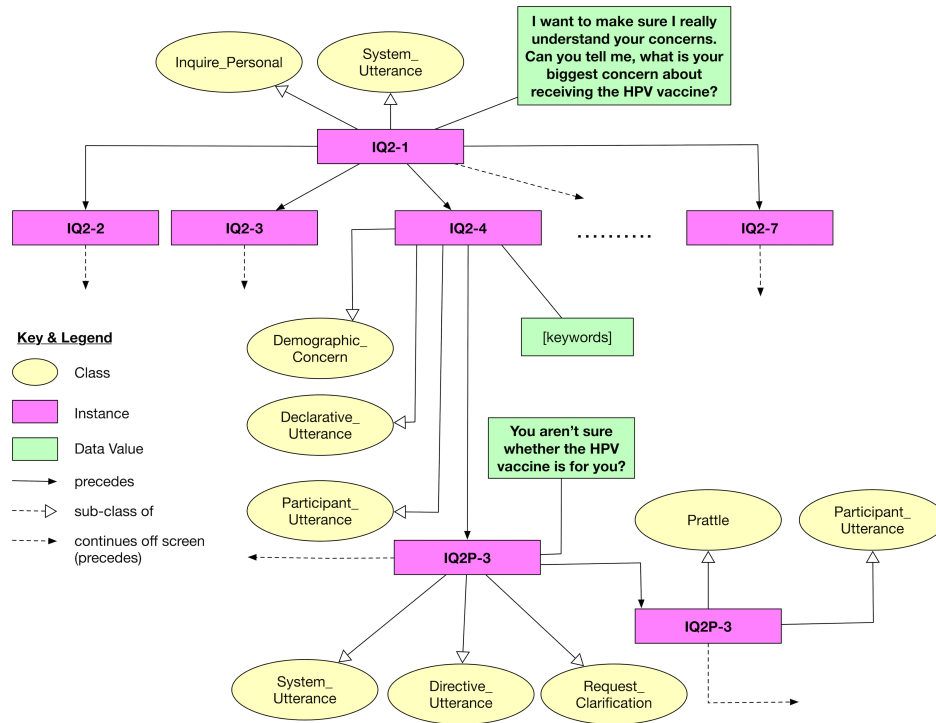


Figure 2: An example of how the utterance data (instances) structured by the PHIDO-derived MI model. Each utterance instance data is linked with *precedes* object property.

figure, we assigned unique identifiers and their actual utterance string spoken by the machine as a data value. If the utterance is a system utterance the a placeholder is used.

For each subsection of the dialogue model we produced 14 ontology artifacts for review. Appendix Figure 6 outlines the directed connections between the speech tasks. These subsections (i.e., speech tasks) of the dialogue included:

- **Introduction (Engage)** : the provider agent engages the user participant with introduction and initiating the session.
- **Introduction (Vaccine Status)**: the provider agent inquires about the vaccine status of the user participant
- **Interview (Interest in the HPV Vaccine)** : the provider agent gauges the user participant's HPV vaccine perspective
- **Interview (Inquiry about specific concern)**: the provider agent gauges and invites the user participant to share their concerns about the HPV vaccine
- **Interview (Safety)**: the participant agent gauges and invite the user participant to discuss their concerns about the safety of the HPV vaccine, related to perceived barriers of the health belief model
- **Interview (Efficacy)**: the participant agent gauges and invite the user participant to share their concerns about the effectiveness of the HPV vaccine, related to perceived benefits of the health belief model
- **Interview (Demographics)**: the participant agent gauges and invite the user to discuss whether the vaccine is appropriate

for themselves based on demographic association, related to the health belief model's perceived susceptibility.

- **Interview (Accessibility)**: the provider agent gauges and allows the users to discuss whether the HPV vaccine is available to the user, related to the health belief models' perceived barriers.
- **Interview (Trust)**: the provider agent gauges the user's trust in the intitutions (providers, manufacturers, science research) that provide the HPV vaccine, related to the perceived barriers of the health belief model.
- **Interview (Personal Confidence)**: the provider agent asks the user to rate their own confidence in the HPV vaccine, related to the perceived barriers of the health belief model.
- **Interview (Assess Readiness)**: the provider agent asks the user to rate their own willingness to seek out the HPV vaccine, related to perceived self-efficacy.
- **Inform (Ask-Offer-Ask v.1)**: the provider agents inquires whether the participant user wants any additional information about a specific topic.
- **Inform (Ask-Offer-Ask v.2)**: this is similar to v.1 but provides variant expression of the same utterances.
- **Conclusion**: the provider agent concludes the session and highlights the personal decision to seek out the HPV vaccine.

Each of the utterances for the above mentioned section were translated in OWL2 with their utterances of the participant and the system linked together to form network structure of the dialogue exchanges.

3 RESULTS AND DISCUSSION

We created an ontology artifact (Figure 3) for 14 essential speech tasks from our dialogue model of MI for HPV vaccine uptake. All of the dialogue utterances were encoded as instance data that is limited to specific utterances types in PHIDO including newly created classes specific to motivational interviewing (*Reflective Listening*, *Affirmation*, etc.). In total, we produced 51 new classes for the PHIDO framework and 247 instance of utterances. In addition each of the instances were linked together to cover the entire dialogue flow of our model. Our work is posted on GitHub [25] with the GPLv3 license. With the resulting encoded ontology of MI speech tasks, we can execute the interaction between the user and the agent/machine. Along with the computable representation, we also provide the documented model/script under GPLv3 at our GitHub repository.

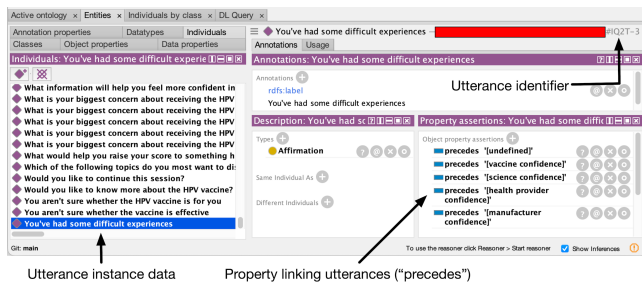


Figure 3: Screenshot of the ontology in Protégé [28] with annotations showing utterance instances, property links and identifiers

We developed a dialogue model to guide conversational agents to perform MI to encourage uptake of the human papillomavirus vaccine targeting young adults. The model utilizes key MI elements, such as OARS statements, asking permission, asking to weigh pros and cons, and a readiness scale. It uses a framework of TTM to move the user through the stages of change, and identifies elements of the HBM that may be influencing the user's readiness for behavior change. In addition, we developed a computable representation of our dialogue model to enable machines to have the knowledge and ability to perform MI counseling specific to HPV vaccine. The ontology uses the PHIDO framework to help provide foundational building blocks to create our ontology. In our previous work, we demonstrated the use of PHIDO in several previous studies to direct machines in interacting with human users. This is assisted by a dialogue engine that uses PHIDO-based ontology.

With the completed model, we intend to work with our collaborators to test the dialogue model in a Wizard of Oz experiment with live participants. The Wizard of Oz experiment entails having live participants interact with a natural language interface of the conversational agent with an operator feeding the agent utterances from the script [14]. The purpose of this experiment will provide early insight on how users might respond to a hypothetical automated conversational agent (i.e., usability). In addition, this experiment will also help evaluate the quality of the dialogue model by capturing situations not accounted for, and possible revision of our dialogue model. Earlier, we noted others have experimented with conversational agents in HPV vaccine counseling using a Wizard of

Oz experiment [4, 7]. Presumably, adding motivational interviewing could show increased positive outcomes if we were to measure change in attitudes toward the HPV vaccine and examine future HPV vaccine uptake of the participants.

A challenge with this work was manageability of the dialogue model. While the detailed visualization of the dialogue model helped to structure the interaction, performing modifications, constraints of the visual notation, and eventual manifestation of the computable version were time consuming. However, we foresee the possibility of developing tool that can facilitate the management and production of the dialogue model. Some of the processing pipeline described in Section 2.2.2 have some of the foundational method developed and can be repackaged for a tool.

The dialogue model was restricted to the English language, and may not be easily translatable to a non-native English speaker. This is attributed due to cultural differences and possible attitudes that may vary with different segments of patient populations. For example, attitudes toward sexual behaviors (a behavior that contribute to the spread of HPV) may be varied due to cultural norms and acceptance [20, 26]. Additionally, the dialogue model focused on a young adult population who may have missed the HPV vaccine due to its availability and potential freedom to make their own health decisions. Thus, the dialogue model may need to be tailored for a parental patient audience if we are to direct our efforts to parents with children of age to take the HPV vaccine.

Another limitation of this work is measuring fidelity to motivational interviewing. For practitioners of motivational interviewing, there exist a fidelity scale to assess a subject's performance in administering MI to a patient [27]. This evaluation is conducted by MI-trained professionals, a resource we currently lack to assess (co-author JH is a MI-trained professional who helped developed the dialogue model). A future step will include a group of MI experts using the fidelity scale to assess the dialogue model's adherence to MI principles – this would indicate probable effectiveness to use MI for a specific goal.

4 CONCLUSION

This paper describes our initiative to transform a structured counseling conversational method, motivational interviewing, into an ontology-based model that can direct agent-based machines to autonomously operate with human users. The contributions of our effort is that we used an ontological based method (based on an existing ontology framework) to represent MI-based dialogue specific to HPV vaccine encouragement for young adult patients in the clinic. Another contribution of this work is that we have also created a dialogue model that we can later use for training or simulation purposes, such as the aforementioned Wizard of Oz experiment in a clinical testing environment. Our computable ontology model is available on our GitHub repository. Future work will involve assessing the dialogue model with MI experts, testing and augmenting the ontology for a dialogue engine, and re-purposing the dialogue model for training purposes for clinician students.

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APPENDIX: FIGURES

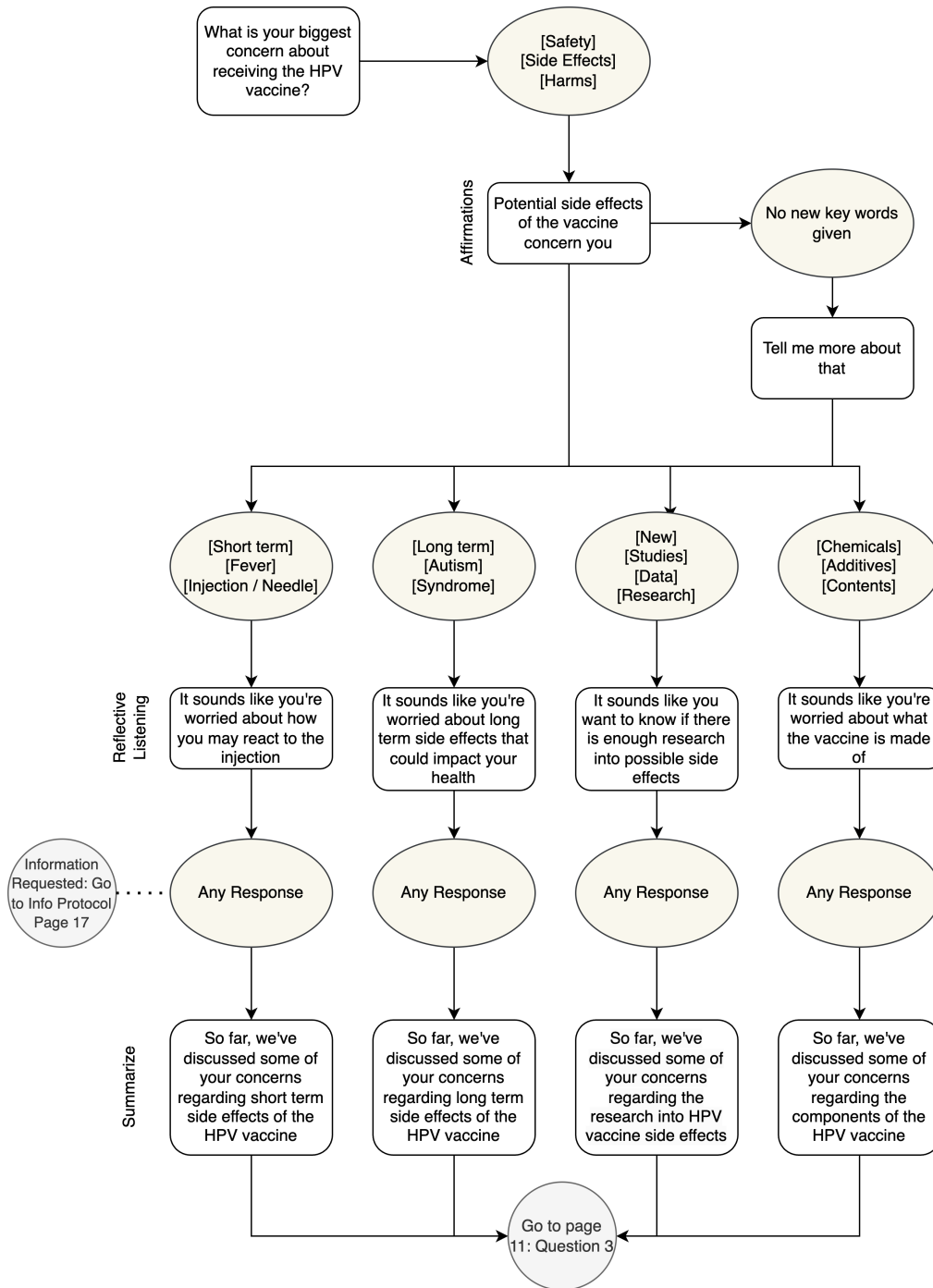


Figure 4: This sample of the script demonstrates the OARS approach of the interview phase, beginning with an open-ended question, followed by affirmation, reflective listening, and summarizing. This framework is utilized for every question in the interview phase.

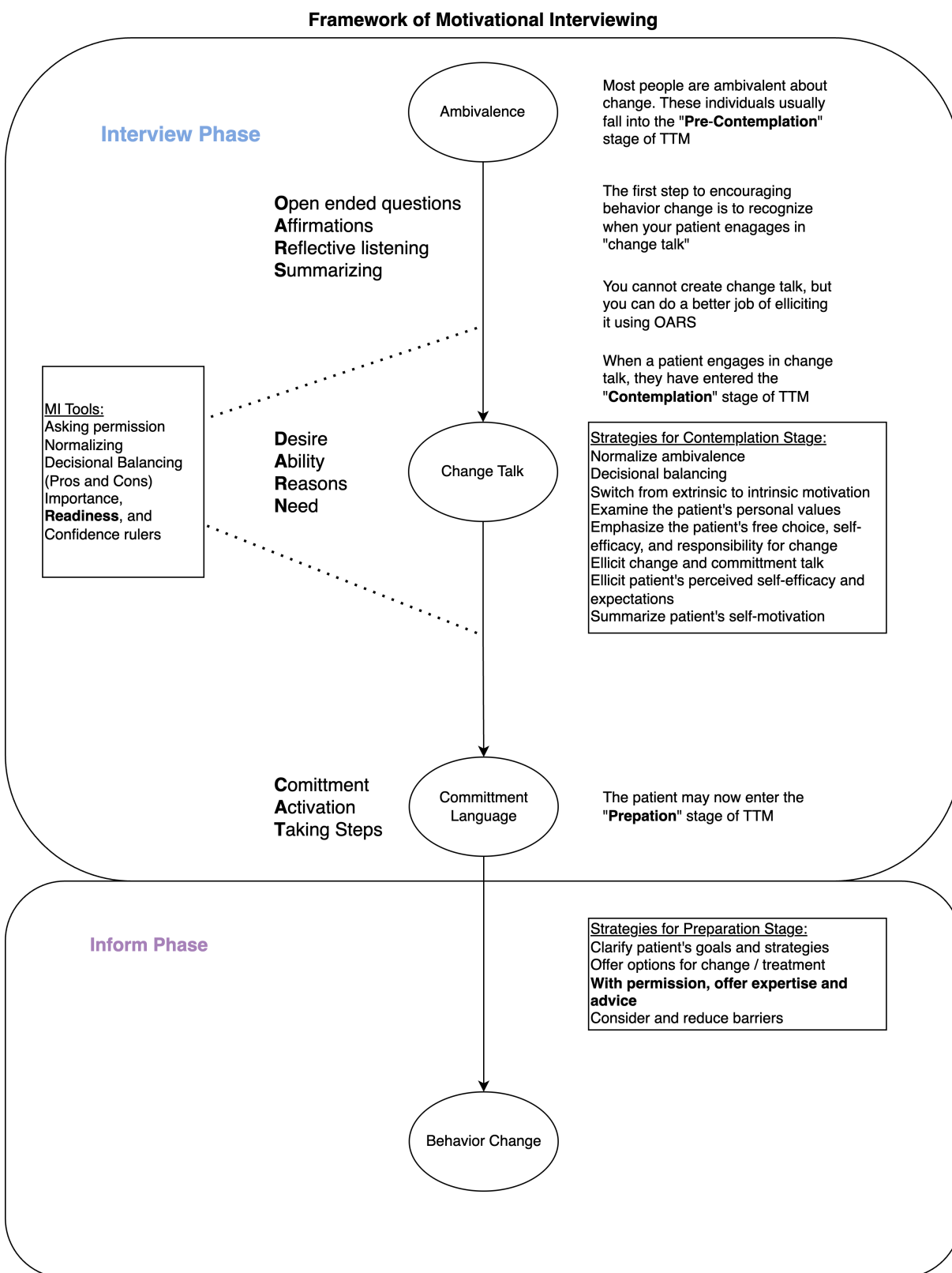


Figure 5: Outline of the model's framework, with application of MI and TTM principles.

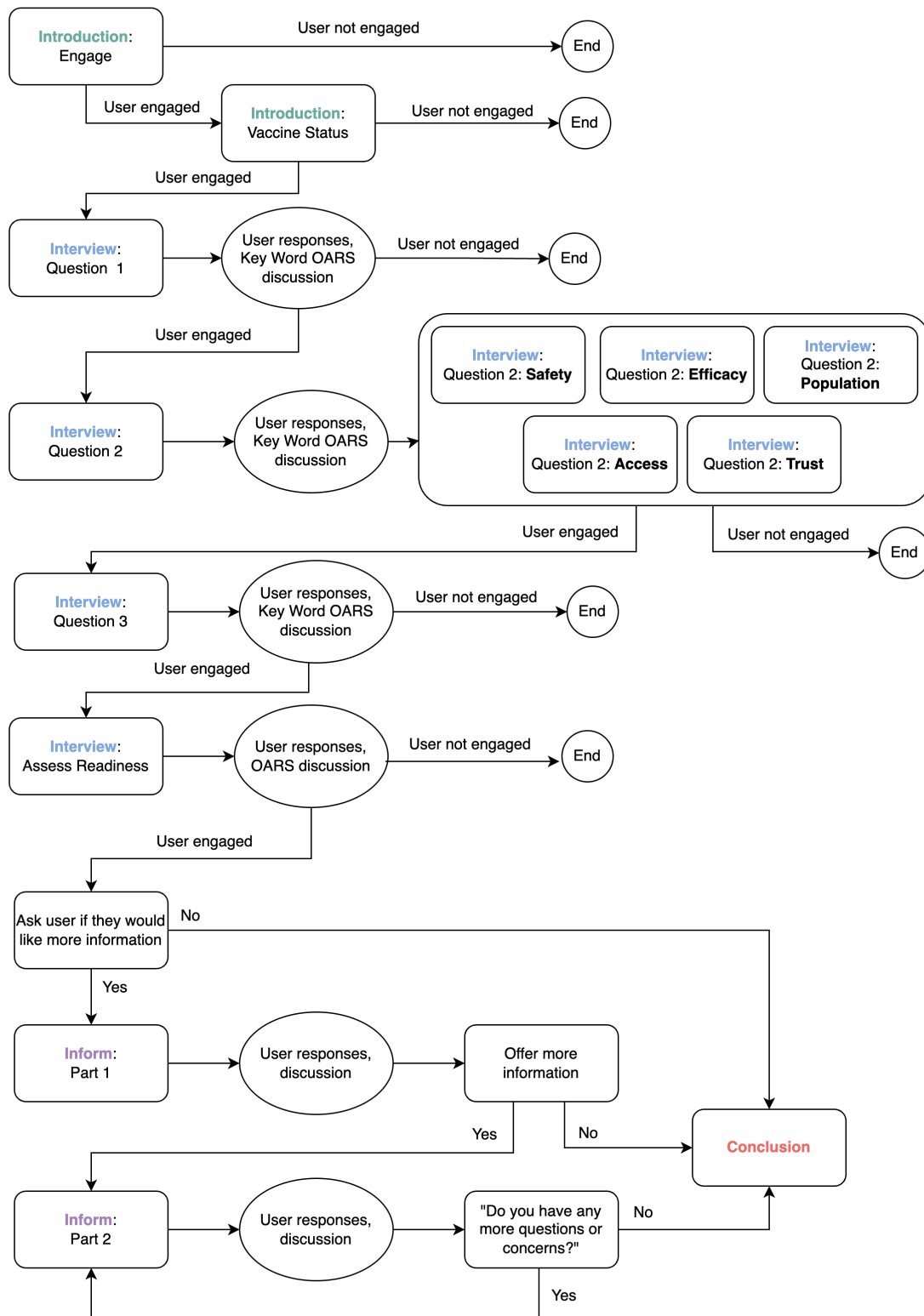


Figure 6: Overall outline of the main speech tasks of the dialogue model. Full version of the dialogue model is available at our GitHub repository [25].

APPENDIX: TABLES

Table 1: New utterance types specific to motivational interviewing

Utterance Class	Subtype	SubClass
Accessibility Concern	Declarative_Utterance	Participant_Utterance
Adverse Effects Concern	Declarative_Utterance	Participant_Utterance
Affirmation	Declarative_Utterance	System_Utterance
Affirmative Response	Declarative_Utterance	Participant_Utterance
Age Concern	Demographic Concern	Declarative_Utterance
Assent Response	Declarative_Utterance	Participant_Utterance
Benefit Efficacy Concern	Efficacy Concern	Declarative_Utterance
Compositional Effect Concern	Adverse Effects Concern	Declarative_Utterance
Confidence Concern	Declarative_Utterance	Participant_Utterance
Cost Accessibility Concern	Accessibility Concern	Declarative_Utterance
Demographic Concern	Declarative_Utterance	Participant_Utterance
Demographic Efficacy Concern	Efficacy Concern	Declarative_Utterance
Dissent Response	Declarative_Utterance	Participant_Utterance
Efficacy Concern	Declarative_Utterance	Participant_Utterance
Experimental Effect Concern	Adverse Effects Concern	Declarative_Utterance
Exposure Efficacy Concern	Efficacy Concern	Declarative_Utterance
Gender Concern	Demographic Concern	Declarative_Utterance
General HPV Concern	Declarative_Utterance	Participant_Utterance
Health Accessibility Concern	Accessibility Concern	Declarative_Utterance
Health Condition Concern	Demographic Concern	Declarative_Utterance
Health Provider Confidence Concern	Confidence Concern	Declarative_Utterance
Individual Susceptibility Concern	Demographic Concern	Declarative_Utterance
Inquire Readiness	Directive_Utterance	System_Utterance
Location Accessibility Concern	Accessibility Concern	Declarative_Utterance
Long Term Effect Concern	Adverse Effects Concern	Declarative_Utterance
Manufacturer Confidence Concern	Confidence Concern	Declarative_Utterance
Neutral Desire	Commissive_Utterance	Participant_Utterance
Neutral Readiness Response	Representative_Utterance	Participant_Utterance
Neutral Response	Declarative_Utterance	Participant_Utterance
Offer	Directive_Utterance	System_Utterance
Permission to Continue Session	Directive_Utterance	System_Utterance
Permission to Discuss Health Information	Directive_Utterance	System_Utterance
Permission to Inquire Personal	Directive_Utterance	System_Utterance
Reflective Listening	Representative_Utterance	System_Utterance
Request Clarification	Directive_Utterance	System_Utterance
Research Efficacy Concern	Efficacy Concern	Representative_Utterance
Science Confidence Concern	Confidence Concern	Representative_Utterance
Short Term Effect Concern	Adverse Effects Concern	Representative_Utterance
Strong Negative Desire	Commissive_Utterance	Participant_Utterance
Strong Negative Readiness Response	Representative_Utterance	Participant_Utterance
Strong Positive Desire	Commissive_Utterance	Participant_Utterance
Strong Positive Readiness Response	Representative_Utterance	Participant_Utterance
Strong Positive Vaccine Intention	Commissive_Utterance	Participant_Utterance
Suggest Question Options	Directive_Utterance	System_Utterance
Summarize	Declarative_Utterance	System_Utterance
Time Accessibility Concern	Accessibility Concern	Representative_Utterance
Vaccine Confidence Concern	Confidence Concern	Representative_Utterance
Weak Negative Desire	Commissive_Utterance	Participant_Utterance
Weak Negative Readiness Response	Representative_Utterance	Participant_Utterance
Weak Positive Desire	Commissive_Utterance	Participant_Utterance
Weak Positive Readiness Response	Representative_Utterance	Participant_Utterance